

# Deep Learning-Enabled Vehicle detection and Traffic Analysis in Indian Urban Mixed Traffic Condition

**A Thesis Report**

*submitted in partial fulfillment of the requirements  
for the award of the degree of*

**Master of Technology**

in

**Civil Engineering**

(Transportation Systems Engineering)

by

**Vivek Rao**

**ROLL NO. 224104517**

Under the supervision of

**Dr. Nipjyoti Bharadwaj**

**Dr. Debanga Raj Neog**

Assistant Professor



**DEPARTMENT OF CIVIL ENGINEERING  
INDIAN INSTITUTE OF TECHNOLOGY GUWAHATI  
ASSAM (INDIA) - 781039**

# DECLARATION

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I hereby certify that the work which is being presented in the M.Tech. Project entitled “**Deep Learning-Enabled Vehicle detection and Traffic Analysis in Indian Urban Mixed Traffic Condition**” in partial fulfillment of the requirements for the award of the **Master of Technology in Civil Engineering (Transportation Systems Engineering)** is an authentic record of my own work carried out during a period from July, 2023 to June, 2024 under the supervision of **Dr. Nipjyoti Bharadwaj, Assistant Professor**, Department of Civil Engineering .

The matter presented in this thesis has not been submitted for the award of any other degree elsewhere.

*Signature of Candidate*

**Vivek Rao**

**Roll No. 224104517**

# CERTIFICATE

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This is to certify that the M.Tech. Project Thesis entitled "**Deep Learning-Enabled Vehicle detection and Traffic Analysis in Indian Urban Mixed Traffic Condition**" submitted by **Vivek Rao (Roll No. 224104517)** is an authentic record of his own work. The project was completed in partial fulfillment of the requirements for the award of the **Master of Technology in Civil Engineering (Transportation Systems Engineering)** and was conducted from July 2023 to June 2024 under my supervision.

Furthermore, the content presented in this thesis has not been submitted for the award of any other degree or qualification in this or any other institution. This statement is correct to the best of my knowledge.

*Signature of Supervisor*

**Date:**

**Dr. Nipjyoti Bharadwaj**  
**Assistant Professor, IITG**

# ACKNOWLEDGEMENT

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First and foremost, I express my gratitude to the Almighty for blessing me with the zeal and enthusiasm to complete this research work successfully.

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Most importantly, I would like to express my heartfelt gratitude to my parents for their constant support and blessings, without which this project would not have been possible.

[Vivek Rao]

# ABSTRACT

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The Research work introduces a deep learning-enabled vehicle detection and traffic analysis system, optimized for the diverse and unstructured traffic conditions typical in Indian urban settings. Utilizing the YOLOv8 (You Only Look Once v8) model, the study addresses the unique challenges posed by Indian city traffic. The methodology encompasses dataset preparation, model training, and deployment for vehicle detection, classification, and traffic flow analysis.

Results reveal high accuracy in identifying and classifying various vehicle types such as motorcycles, cars, buses, and non-motorized vehicles, even amidst mixed traffic conditions. The analysis provides detailed insights into traffic composition, flow, density, and speed compliance, identifying critical patterns for effective traffic management. Key findings highlight the prevalence of motorcycles and cars, significant speed limit non-compliance among motorcycles, and varying levels of service (LOS) influenced by traffic density and flow. Real-time monitoring through the YOLOv8 model offers promising improvements for urban traffic management systems.

However, the model exhibits limitations within a 50-60 meter range, with occasional misclassifications and undetected vehicles. To mitigate these issues and enhance model accuracy, the training dataset needs to be augmented with more diverse and comprehensive images, including more samples of underrepresented classes and additional side-view images. Future research will focus on integrating the system with existing traffic infrastructure, expanding the dataset to improve model robustness, and exploring real-time applications for comprehensive traffic management. Advanced tracking algorithms will be investigated to enhance vehicle trajectory accuracy, with further studies on trajectory prediction and analysis to better understand vehicle movement patterns in mixed traffic.

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# Chapter 1

## INTRODUCTION

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### 1.1 Background

In the wake of rapid urbanization and increasing vehicle ownership, developing countries like India are facing severe traffic congestion, particularly in urban areas. Urban roads are notorious for their traffic jams. This congestion is not merely an inconvenience but a significant contributor to economic losses, environmental pollution, and a reduction in the quality of urban life. Traffic congestion has created unnecessary time consumption and pollutant emissions by using fossil fuels, which has become a serious social problem. The distinctive challenge in these countries arises from the mixed nature of traffic and the absence of structured lane discipline, leading to inefficient road space utilization and heightened congestion.

Traditional traffic management in Indian urban areas predominantly relies on conventional methods. These systems, characterized by their simplicity and cost-effectiveness, do not adapt to real-time traffic variations. While this approach has been the mainstay of traffic control for decades, it is increasingly proving inadequate in managing the complexities of modern urban traffic, especially in the mixed traffic conditions prevalent in Indian cities.

### 1.2 Traffic Dynamics in Indian Urban Areas

#### 1.2.1 Lack of Lane Discipline

Indian roads are often noted for the absence or disregard of lane markings, contributing to indiscipline in lane usage. This leads to a less organized flow of traffic, where vehicles may not adhere strictly to designated lanes. The unpredictable vehicle movements and lane switching complicate traffic management.

### **1.2.2 Diversity of Vehicle Types**

A significant aspect of Indian traffic is the wide spectrum of vehicle types sharing the same road space. From two-wheelers, e-rickshaws, and auto-rickshaws to large trucks and buses, alongside pedestrians, the roads exhibit a diverse and sometimes chaotic mix of traffic.

### **1.2.3 Unstructured Traffic Flow**

Unlike the more homogeneous and regulated traffic patterns seen in many Western countries, Indian roads demonstrate a disorganized blend of various vehicles. This diversity often results in inefficient road space utilization and heightened congestion levels.

### **1.2.4 Heterogeneous Speed Profiles**

Indian roads have a mix of vehicles with varying speed capabilities. From slow-moving cycles and rickshaws to quicker cars and motorbikes, this diversity creates a range of speed profiles within the same traffic flow. These differences in speed lead to frequent braking and accelerating, increasing the chance of traffic jams and accidents, and adding to overall congestion.

### **1.2.5 Informal Pedestrian Movement**

Pedestrian movement in Indian cities often lacks the formal structure of crosswalks or dedicated signals. As a result, pedestrians are compelled to navigate traffic independently, leading to increased unpredictability on the roads. This spontaneous movement of pedestrians requires extra attention from drivers and makes the traffic flow more complex.

### **1.2.6 Presence of Non-Motorized Traffic**

A significant volume of pedestrians and non-motorized vehicles like bicycles and hand carts adds to the complexity of traffic, often moving unpredictably and sharing space with motorized vehicles.

### **1.2.7 Road Infrastructure Constraints**

Road infrastructure in many Indian cities is unable to keep up with the growing number of vehicles. Problems like narrow roads, limited pedestrian paths, insufficient parking, and no specific lanes for different vehicles increase traffic jams and inefficiency, highlighting the need for better urban planning and development.

### **1.2.8 Impact of Street Vendors and Parked Vehicles**

The presence of street vendors and irregular parking, particularly in market areas, often reduces the effective width of roadways. This encroachment not only limits the available space for vehicular movement but also introduces additional elements of unpredictability and disorder in traffic flow. These factors make it harder to manage traffic flow and signal timings, requiring new solutions to handle the challenges of mixed traffic.

## **1.3 Terminology**

### **1.3.1 Types of Traffic: Homogeneous and Mixed Traffic**

#### **Homogeneous Traffic**

Homogeneous traffic is characterized by the presence of vehicles that share similar attributes in terms of size, speed capabilities, and manoeuvrability. Such conditions are often observed in scenarios where specific lanes are designated for specific types of vehicles. This uniformity results in a traffic flow that is considerably more predictable and easier to manage. Homogeneous traffic follows strict lane discipline compared to mixed traffic. The primary advantage of this type of traffic is the relative ease of implementing traffic control and management strategies, as the similar vehicular characteristics lead to a more streamlined flow and reduced complexities in traffic handling.

## **Mixed Traffic**

Mixed traffic refers to a scenario where a variety of vehicle types coexist on the same roadway. This mix typically includes a wide range of sizes (motorcycles, bicycles, cars, buses, trucks, etc.), differing speed capabilities, and varying degrees of manoeuvrability. Also, the absence of strict lane discipline is often evident, leading to irregular positioning and alignment of vehicles on the roadway. Such diversity is predominantly seen in urban and semi-urban road networks. Mixed traffic conditions pose significant challenges in terms of traffic management and safety. The heterogeneous nature of the vehicles necessitates the adoption of more sophisticated and multifaceted traffic engineering solutions.

## **1.4 Vehicle Detection and Classification**

Object detection algorithms are broadly classified into two categories based on how many times the same input image is passed through a network. Object detection algorithms can be divided into two main categories: single-shot detectors and two-stage detectors.

### **1.4.1 Single-Shot Object Detection**

Single-shot object detection uses a single pass of the input image to make predictions about the presence and location of objects in the image. It processes an entire image in a single pass, making it computationally efficient.

### **1.4.2 Two-Stage Object Detection**

Two-stage object detection uses two passes of the input image to make predictions about the presence and location of objects. The first pass is used to generate a set of proposals or potential object locations, and the second pass is used to refine these proposals and make final predictions. Overall, the choice between single-shot and two-shot object detection depends on the specific requirements and constraints of the application. Generally, single-shot object detection is better suited for real-time applications, while two-shot object detection is better for applications where

accuracy is more important.

### 1.4.3 YOLOv8 (You Only Look Once version 8)

YOLO is a state-of-the-art, real-time object detection system. It applies a single neural network to the full image, dividing the image into regions, and predicting bounding boxes and probabilities for each region. The bounding boxes are weighted by the predicted probabilities. YOLOv8 is the first version of YOLO released in 2023. The main idea that makes YOLO so attractive is its ability to detect objects in an image in a single pass. At the time, this was a breakthrough because most algorithms had to be run repeatedly on different parts of the image.

### 1.4.4 Mathematical Foundations of YOLOv8

YOLOv8's detection mechanism can be mathematically described as follows:

1. **Bounding Box Prediction** The bounding box is predicted by a single fully connected layer, which outputs:

$$(b_x, b_y, b_w, b_h, c)$$

where  $b_x$  and  $b_y$  represent the center coordinates of the box,  $b_w$  and  $b_h$  are the width and height, and  $c$  is the confidence score.

2. **Loss Function** YOLOv8 uses a multi-part loss function to optimize bounding box predictions and class probabilities simultaneously:

$$L = L_{\text{coord}} + L_{\text{conf}} + L_{\text{class}}$$

where  $L_{\text{coord}}$  penalizes errors in bounding box coordinates,  $L_{\text{conf}}$  penalizes errors in object confidence scores, and  $L_{\text{class}}$  penalizes errors in class prediction.

3. **Intersection over Union (IoU)** The IoU metric is used to evaluate the accuracy of the predicted bounding boxes:

$$\text{IoU} = \frac{A \cap B}{A \cup B}$$

where  $A$  and  $B$  are the predicted and ground truth bounding boxes, respectively.

### 1.4.5 Benefits of YOLOv8 for Traffic Analysis

YOLOv8 offers several advantages that make it particularly suitable for traffic analysis in urban environments:

- **Real-Time Processing:** YOLOv8's architecture is optimized for real-time processing, allowing for the continuous monitoring of traffic conditions with minimal latency.
- **High Accuracy:** The model's enhanced feature extraction capabilities lead to higher detection accuracy, crucial for identifying various vehicle types in mixed traffic conditions.
- **Scalability:** YOLOv8 can be scaled to run on different hardware configurations, from powerful GPUs in centralized traffic management centers to edge devices deployed at traffic intersections.
- **Robustness to Variability:** The anchor-free approach and advanced feature aggregation make YOLOv8 robust to the variability in vehicle sizes, shapes, and occlusions commonly encountered in urban traffic.

### 1.4.6 Applications of YOLOv8 in Traffic Analysis

YOLOv8 can be applied to various aspects of traffic analysis, enhancing the capabilities of urban traffic management systems:

- **Vehicle Detection and Classification:** YOLOv8 can accurately detect and classify different types of vehicles, including cars, trucks, motorcycles, and buses, even in congested and heterogeneous traffic environments.
- **Traffic Density Estimation:** By analyzing the number of vehicles and their spatial distribution, YOLOv8 can provide real-time estimates of traffic density, aiding in congestion management and road planning.
- **Traffic Flow Analysis:** The model can track vehicle movements across multiple frames, enabling the analysis of traffic flow patterns and identification of bottlenecks.

- **Incident Detection:** YOLOv8 can be used to detect traffic incidents such as accidents, stalled vehicles, and illegal parking, facilitating quicker response and resolution by traffic authorities.
- **Pedestrian Detection:** In addition to vehicles, YOLOv8 can also detect pedestrians, improving safety measures at crosswalks and intersections.

## 1.5 Motivation

The motivation behind this research stems from the pressing need to address the inefficiencies in current traffic management systems in Indian urban settings. The distinctive challenges posed by mixed traffic, characterized by the coexistence of various vehicle types, lack of lane discipline, and unstructured traffic flow, necessitate the development of advanced, adaptive solutions. The integration of deep learning techniques into traffic management can provide real-time data, improve traffic signal optimization, and ultimately enhance the efficiency and effectiveness of urban traffic management.

## 1.6 Scope

This research aims to develop and deploy a deep learning-enabled vehicle detection and traffic analysis system tailored for Indian urban mixed traffic conditions. The scope includes the collection and analysis of traffic data from diverse urban settings, the training and validation of a YOLO (You Only Look Once) deep learning model for vehicle detection, and the evaluation of traffic flow dynamics. By focusing on the unique characteristics of Indian traffic, this study seeks to provide insights into congestion patterns, vehicle speed compliance, and overall traffic management improvements.

# Chapter 2

## LITERATURE REVIEW

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In addressing the challenges of vehicle detection and traffic analysis in urban mixed traffic conditions, a comprehensive review of the existing literature is indispensable. This study explores the integration of deep learning techniques, specifically YOLOv8, in enhancing vehicle detection and traffic analysis in Indian urban settings, which are characterized by heterogeneous and often chaotic traffic patterns. The papers reviewed in this section provide a broad spectrum of methodologies and technologies employed in traffic congestion analysis, vehicle detection, and traffic management systems.

### **Sheng et al. (2024)**

Sheng et al. (2024) introduced the YOFGD, a multi-objective pedestrian tracking method combining YOLOv8 and an enhanced DeepSORT, targeting local occlusion and ID dynamic transformation issues in traffic scenarios. The integration of YOLOv8 for high-level small-scale feature extraction and OSNet within DeepSORT improved real-time target tracking and edge recognition. The model, using OSNet, FSA, and GFModel, achieved an accuracy of 77.9% and a speed of 55.8 FPS, demonstrating its practical applicability.

### **Soma et al. (2024)**

Soma et al. (2024) tackled vehicle classification and localization challenges for Intelligent Transport Systems (ITS) using YOLOv8. The study focused on urban environments with closely packed vehicles and image occlusions. YOLOv8's real-time detection and vehicle speed calibration via pixel-meter conversion improved the system's accuracy and precision in traffic flow analysis, safety planning, and automated toll systems. The model achieved a mean average precision of 54%, outperforming previous algorithms.

**Telambauna et al. (2023)**

Telambauna et al. (2023) employed YOLOv8 for detecting nine vehicle classes in an intersection dataset, achieving 77% accuracy with 100 epochs, a batch size of 8, and a 640 image size. Despite successful vehicle type detection, occlusion issues reduced accuracy, highlighting the need for addressing overlapping vehicle objects [? ].

**Wang et al. (2024)**

Wang et al. (2024) enhanced the YOLOv8n model to address large-scale variations, computational demands, and storage issues in transportation target detection. By embedding the BiFormer attention mechanism in the Neck layer, adding a small-scale target detection header, and using a weighted bidirectional feature pyramid structure, the model's mAP improved by 4.7 points to 95.9%, and mAP50:95 by 6.2 points to 74.5% .

**Agarwal et al. (2024)**

Agarwal et al. (2024) developed the Indian Traffic Dataset (ITD) to address existing datasets' limitations in handling diverse vehicle types. ITD includes 17,666 annotated images and 341,000 objects, suitable for South and Southeast Asian traffic studies. The dataset showed no significant difference between observed and predicted traffic volumes, validating its reliability .

**Ilamati et al. (2024)**

Ilamati et al. (2024) proposed the HL-MVD framework using CMATDet for multi-vehicle detection in unregulated traffic. The framework used a Pool-based Active Learning Strategy, transfer learning with YOLOv5x, and a Multi-head Attention transformer encoder. It achieved high mAP scores and lower inference latency, outperforming recent detection models .

**Aswathi et al. (2022)**

Aswathi et al. (2022) designed a traffic congestion analysis system using YOLO for real-time vehicle detection. Videos from Bangalore's skywalks and other sources

trained the system, achieving recognition accuracies between 76.70%-89.24% with YOLOv3 and 83.08%-92.65% with YOLOv4. The system provided comprehensive congestion analysis reports.

**Yong et al. (2024)**

Yong et al. (2024) studied vehicle behavior at intersections using YOLOV7 and DeepSORT, analyzing vehicle speed and turning angles. The research provided optimal turning speeds and angles, contributing to road safety measures.

**Bekele et al. (2022)**

Bekele et al. (2022) utilized one-stage object detectors to detect traffic congestion in Ethiopia. YOLOv3 and YOLOv5 models, trained on manually labeled frames, achieved 68.6% and 62.3% mAP, respectively, highlighting the method's efficiency in real-time traffic analysis.

**Chakraborty et al. (2018)**

Chakraborty et al. (2018) used YOLO and DCNN for detecting traffic congestion from CCTV images, achieving 91.5% and 90.2% accuracy, respectively. The study emphasized deep learning's robustness even under poor camera conditions.

**Hema et al. (2024)**

Hema et al. (2024) proposed a multi-sensor system integrating RFID sensors, cameras, and microphones to improve ambulance navigation through traffic. The system dynamically adjusted traffic signals and routes based on real-time congestion data, enhancing emergency response .

**Huu et al. (2023)**

Huu et al. (2023) developed a system using YOLOv8 and DeepSORT to detect vehicles and calculate traffic density, providing real-time congestion notifications via Telegram. The system achieved 92.9% accuracy and offered comprehensive traffic management information .

**Ifthikar et al. (2023)**

Ifthikar et al. (2023) reviewed DL approaches for UAV-based target detection in traffic congestion scenarios. The study compared one-stage and two-stage detectors, emphasizing accuracy and computational efficiency improvements for urban traffic analysis .

**Khalladi et al. (2024)**

Khalladi et al. (2024) proposed a microscopic approach for categorizing traffic flow into light, medium, and heavy classes using YOLOv5 and vehicle tracking. The approach, tested with various classifiers, achieved 100% accuracy and reduced computational load, enhancing traffic state estimation systems.

**Kumar et al. (2023)**

Kumar et al. (2023) used surrogate safety measures to assess traffic capacity and safety at unsignalized intersections, designing appropriate traffic signals and employing 3D simulation models to improve driver safety .

**Yang et al. (2021)**

Yang et al. (2021) developed a deep learning-based traffic state discrimination method using YOLOv3 and LK optical flow for vehicle speed detection. The algorithm accurately judged traffic states, demonstrating strong anti-interference ability .

## **2.1 Need of Study**

The YOLO (You Only Look Once) model is widely utilized in object detection tasks and is commonly trained on the COCO (Common Objects in Context) dataset. This dataset encompasses a substantial number of object classes, allowing YOLO to detect and identify a diverse array of objects. Despite its extensive class repertoire, the COCO dataset does not include all possible object classes. This limitation becomes particularly evident in the context of traffic studies conducted under Indian conditions, where the variety of vehicle classes present differs significantly from those included in the COCO dataset.

Consequently, the standard YOLO model trained on COCO lacks the ability to accurately detect and identify many vehicle types specific to Indian traffic scenarios. This poses a significant challenge for researchers aiming to conduct precise and comprehensive traffic studies in these conditions. The inability to recognize these unique vehicle classes leads to gaps in data collection and analysis, ultimately impacting the accuracy and applicability of research findings.

Traffic studies typically encompass a wide range of analyses, including vehicle counting, traffic flow analysis, congestion detection, speed estimation, and incident detection. These studies are crucial for understanding traffic patterns, optimizing traffic signal timings, improving road safety, and planning infrastructure development. Accurate vehicle detection and classification are foundational to these studies, as they provide essential data on traffic volume, vehicle types, and movement patterns.

Addressing this gap is imperative for enhancing the precision and relevance of traffic studies in Indian contexts. By training the YOLO model to include the specific vehicle classes prevalent in India, researchers can achieve more accurate object detection and data analysis. This study is therefore essential, as it aims to modify and retrain the YOLO model to encompass a broader range of vehicle classes, tailored to the unique traffic conditions found in India. Through this enhancement, the study seeks to facilitate more accurate, relevant, and actionable insights in the field of traffic research in India, thereby contributing to the development of more effective traffic management and policy-making.

## **2.2 Aim and Objectives**

### **2.2.1 Aim**

The primary aim of this research is to develop a robust, real-time vehicle detection and traffic analysis system using deep learning techniques, specifically tailored for Indian urban mixed traffic conditions.

### **2.2.2 Objectives**

1. To train and validate a YOLO deep learning model for accurate vehicle detection, classification, and counting in Indian traffic conditions.
2. To analyze and quantify traffic flow, density, and speed using the outputs from the YOLO model to gain insights into traffic dynamics in Indian urban conditions.
3. To estimate and evaluate the congestion and level of service (LOS) on Indian urban roads.
4. To identify and analyze speed compliance using the trained YOLO model.
5. To explore the scalability and adaptability of the YOLO model for use in Indian traffic conditions.

# Chapter 3

## DATA COLLECTION AND METHODOLOGY

---

### 3.1 Methodology

A detailed methodology is essential to achieve the objectives outlined in this study. The following sections describe the steps involved in the project.

#### 3.1.1 Project Flowchart

The project flowchart outlines the sequence of steps from data collection to analysis and interpretation.

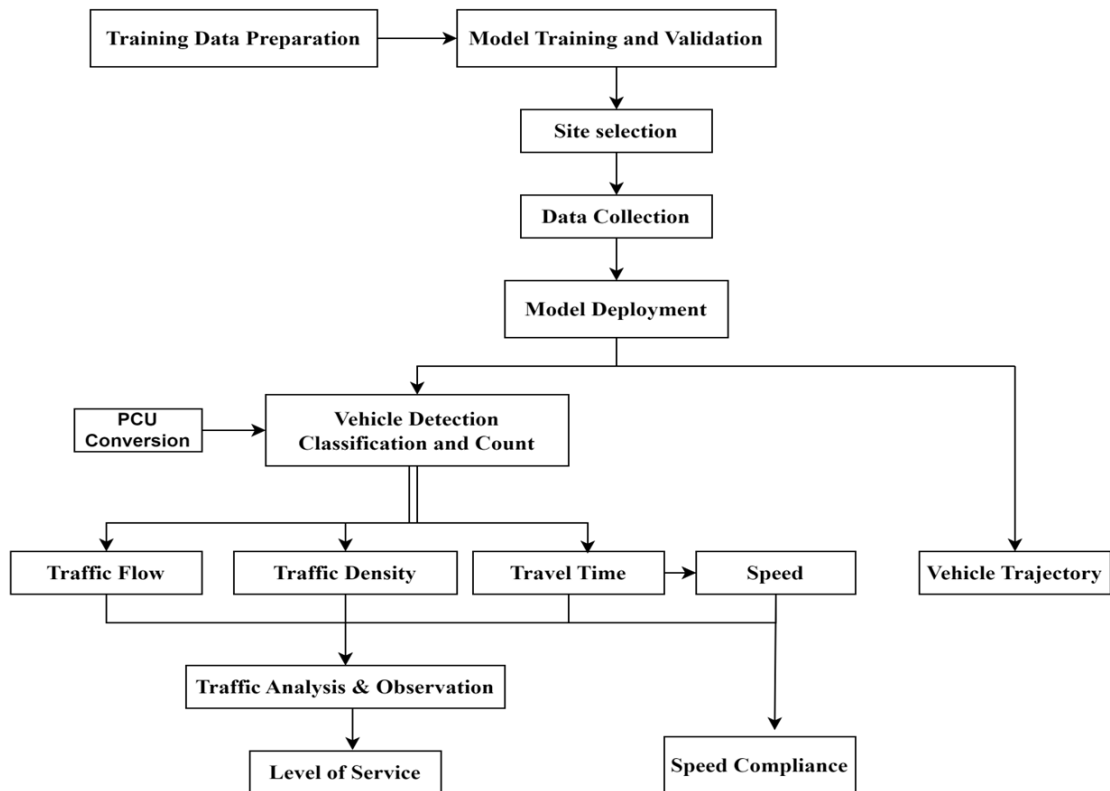


Figure 3.1: Flow chart of Methodology

### 3.1.2 Training Dataset Preparation

The preparation of the training dataset involved a systematic and comprehensive process aimed at ensuring diversity and accuracy in the collected data. This process commenced with the collection of traffic videos from three distinct locations: GS Road in Guwahati, Centre Point T-junction in North Guwahati, and a highway to capture a variety of vehicle classes. These locations were chosen to provide a broad representation of mixed traffic scenarios prevalent in Indian cities.

#### Data Collection and Preprocessing

1. **Video Collection:** Traffic videos were recorded from the three different locations at various angles, including front, back, and side views. This approach was crucial to capture different perspectives and enhance the detection accuracy of various vehicle types.
2. **Frame Extraction:** The collected videos were then converted into individual frames. To avoid redundancy, frames were extracted at specific intervals to ensure that the same vehicle does not appear multiple times, thereby maintaining the diversity of the dataset.
3. **Data Cleaning:** The extracted frames underwent a rigorous data cleaning process, which included removing blurry or low-quality images, as well as duplicate images and frames containing an excessive number of common vehicle types like cars and motorcycles. Once a sufficient count of cars and motorcycles was reached, additional images primarily focused on less common vehicle classes were retained for annotation.
4. **Annotation:** Each image was meticulously annotated to identify and classify different types of vehicles and pedestrians. The annotation process involved labeling the following classes: car, bus, motorcycle, non-motorized vehicle (NMC), pedestrian, truck, light commercial vehicle (LCV), and auto-rickshaw (Auto). This task was carried out using annotation tools that allow precise bounding box placement around the objects of interest.

## Dataset Division

1. **Training Set:** The majority (around 90%, or 4,500 images) of the annotated images were allocated to the training set. This set was used to train the YOLO (You Only Look Once) model, allowing it to learn to detect and classify different vehicles and pedestrians accurately.
2. **Validation Set:** A smaller portion (around 10%, or 500 images) of the annotated images was set aside as the validation set. This set was used to evaluate the model's performance during the training process and to fine-tune the model's parameters to avoid overfitting.

### 3.1.3 Model Training

The YOLOv8l model was trained using the annotated dataset. The training process was executed on an Nvidia GeForce RTX 5090 GPU, which has 16GB of GDDR6 memory. The initial plan was to run the training for 300 epochs, but the process stopped at around 170 epochs. The entire training and validation process took approximately 14 hours. Below are the detailed results and analysis from the training and validation phases.

**Training Results** The results included various performance metrics and plots, such as the confusion matrix, normalized confusion matrix, F1-confidence curve, precision-confidence curve, recall-confidence curve, and other relevant metrics.

#### Confusion Matrix

1. **Standard Confusion Matrix:** Shows the number of true positive, false positive, and false negative predictions for each class.
2. **Normalized Confusion Matrix:** Presents the same data but normalized, which helps in understanding the proportion of correctly and incorrectly classified instances relative to each class.

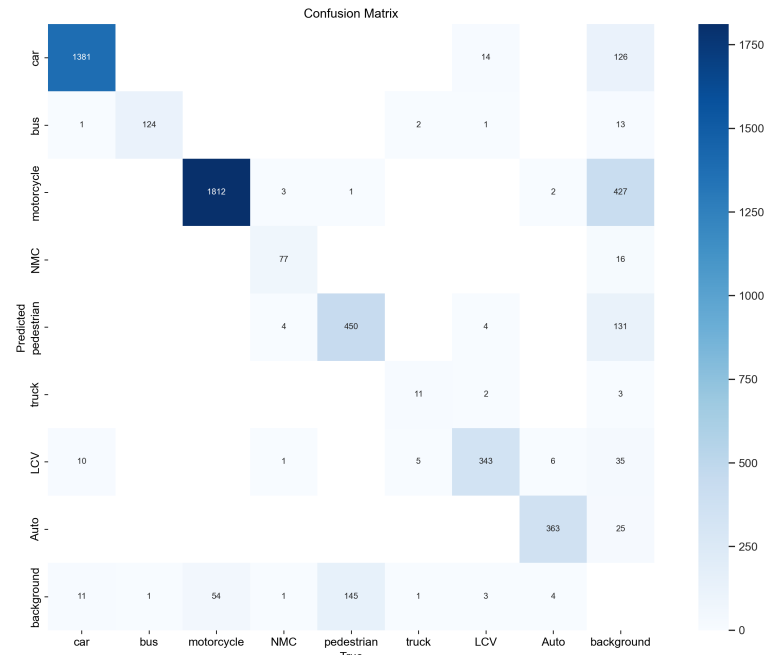


Figure 3.2: Standard Confusion Matrix

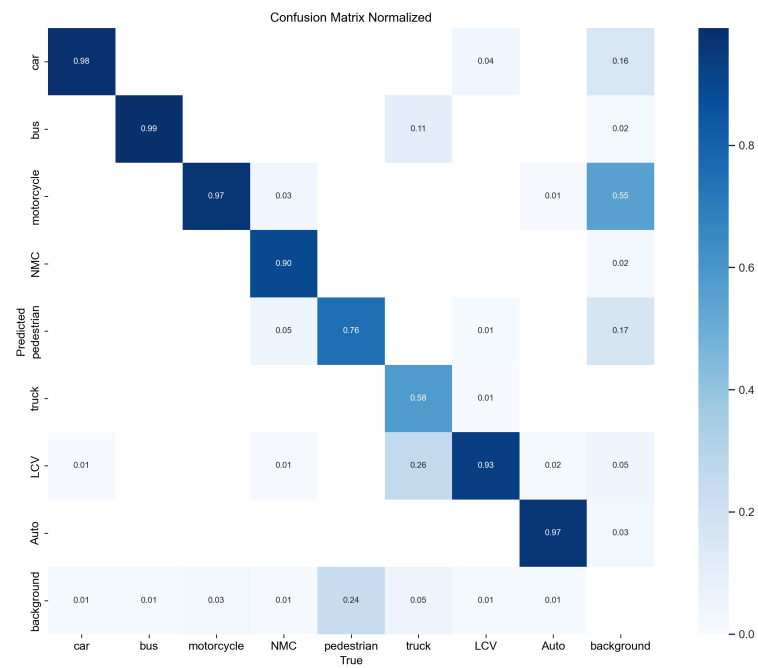


Figure 3.3: Standard Confusion Matrix

## F1-Confidence Curve

1. Illustrates the relationship between the F1 score and confidence levels across different classes.

2. **F1 Score:** A measure combining both precision and recall, providing a single metric to evaluate model performance.

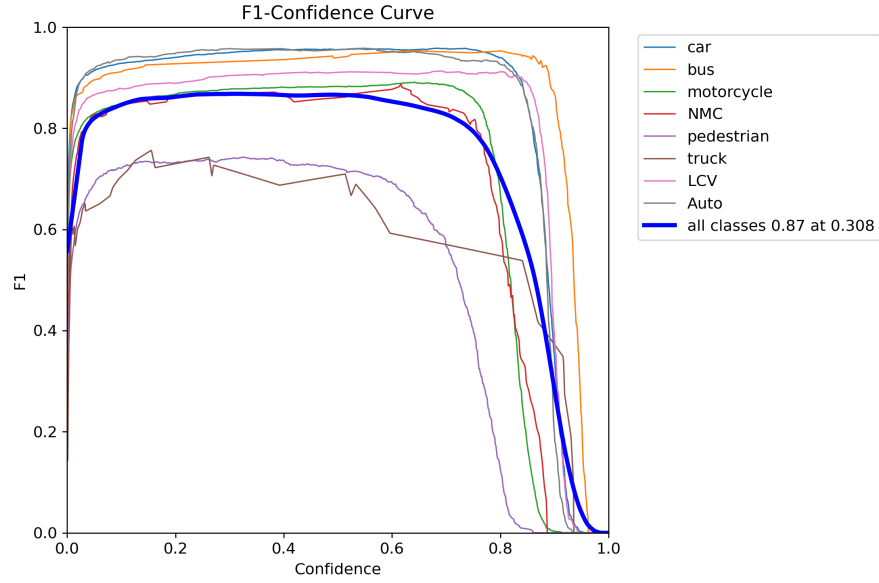


Figure 3.4: F1-Confidence Curve

### Precision-Confidence Curve

1. Demonstrates how precision varies with confidence levels for each class.
2. **Precision:** The ratio of true positive predictions to the total predicted positives (true positives + false positives).

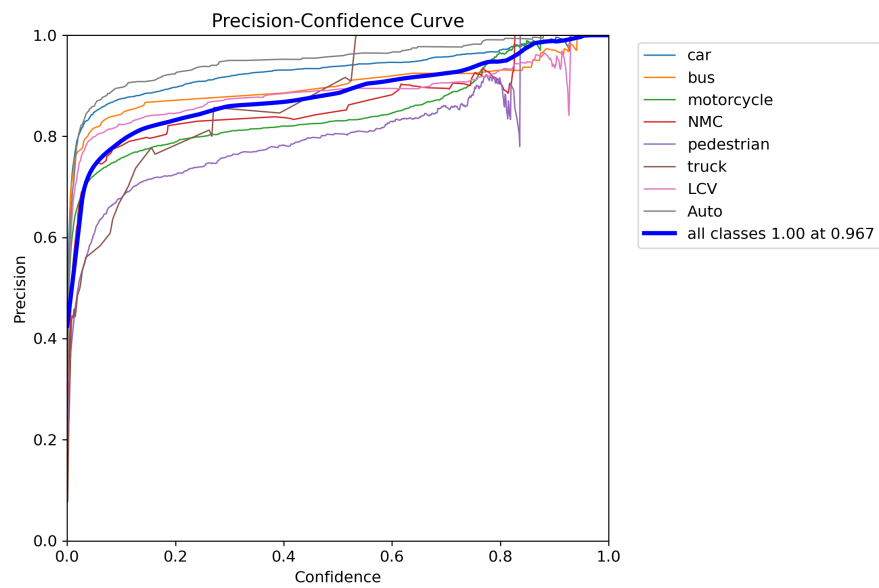


Figure 3.5: Precision-Confidence Curve

## Recall-Confidence Curve

1. Shows how recall changes with different confidence thresholds for each class.
2. **Recall:** The ratio of true positive predictions to the total actual positives (true positives + false negatives).

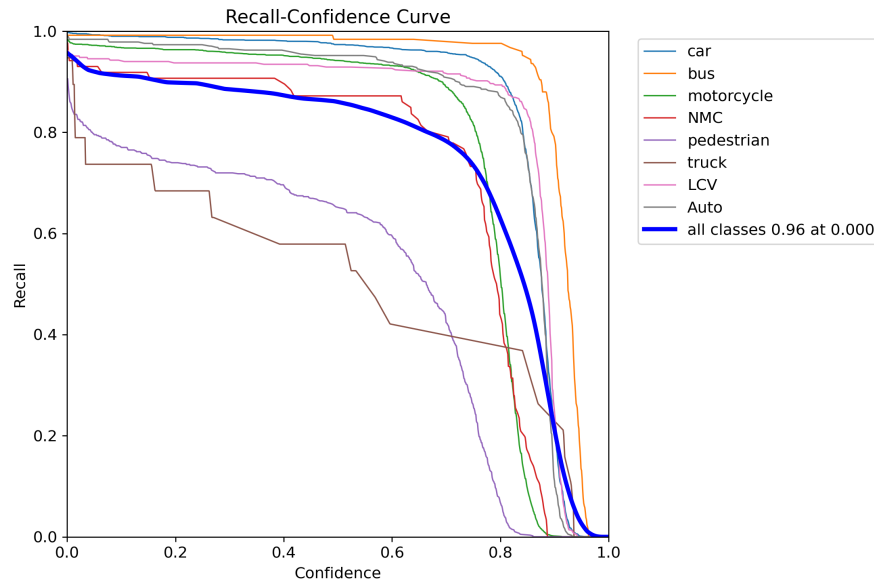


Figure 3.6: Recall-Confidence Curve

## Precision-Recall (PR) Curve

1. Provides a comprehensive view of the trade-off between precision and recall across various confidence levels.
2. **Mean Average Precision (mAP):** A common metric used to evaluate object detection models, representing the average of the precision scores at different recall levels.

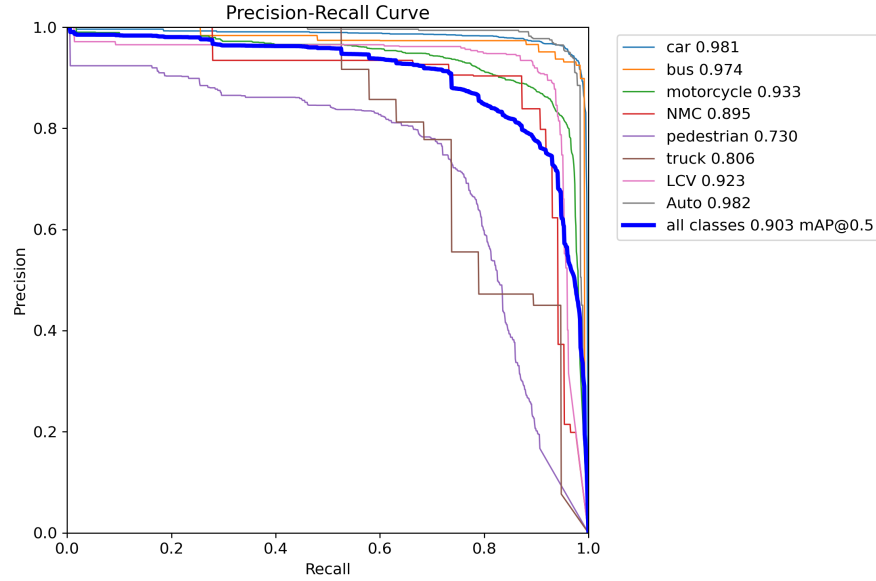


Figure 3.7: Precision-Recall (PR) Curve

## Summary of Performance

### 1. Overall Performance:

- The model demonstrated high precision and recall for most classes, indicating effective learning and accurate predictions.
- **Mean Average Precision (mAP@0.5):** 0.903, showing that the model is highly precise at a 50% IoU threshold.
- **F1 Score:** 0.87, reflecting a balanced trade-off between precision and recall.

### 2. Class-wise Performance:

- **Highest Performance:** Cars and motorcycles showed the highest precision and recall.
- **Improvement Needed:** Classes such as pedestrians and trucks showed lower performance, indicating potential areas for further data augmentation and model tuning.

The YOLOv8l model trained on this dataset is robust and capable of accurately detecting and classifying various vehicle types in mixed traffic conditions.

Further fine-tuning and additional data collection could enhance the model's performance, particularly for the classes with lower precision and recall scores.

### **3.1.4 Comparison Between Trained Model and Pre-trained Model**

The performance of the trained YOLOv8l model was compared with a pre-trained YOLO model to evaluate its accuracy and efficiency in detecting vehicles in Indian traffic conditions. The unique challenges of Indian urban traffic, characterized by a diverse mix of vehicle types, necessitate this comparison to highlight the benefits of a custom-trained model.

#### **Specific Challenges in Indian Urban Traffic**

1. **Diverse Vehicle Types:** Indian urban traffic includes a wide range of vehicles such as thela (handcarts), cycle rickshaws, auto-rickshaws, e-rickshaws, and unique lorry designs. These vehicles vary significantly in shape, size, and speed compared to those typically found in developed countries.
2. **High Proportion of Two-Wheelers:** Two-wheelers, including motorcycles and scooters, constitute a significant portion of the traffic, requiring specialized detection capabilities.
3. **Mixed Traffic Conditions:** The absence of strict lane discipline and the presence of non-motorized vehicles add complexity to vehicle detection and traffic analysis.

Given these factors, the custom-trained YOLOv8l model was developed to cater specifically to Indian urban traffic scenarios.

## Comparison

### 1. Custom-trained YOLOv8l Model:

- **Strengths:** Superior precision, recall, F1 score, and mAP@0.5 for Indian traffic scenarios.
- **Weaknesses:** Slightly higher inference time due to the complexity of handling diverse vehicle types.

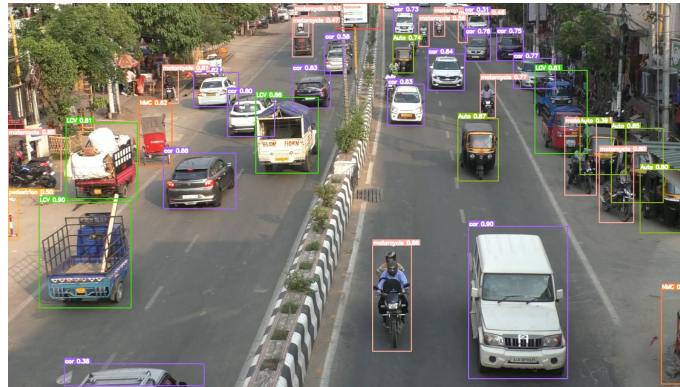


Figure 3.8: Detections and Classifications as per custom model

### 2. Pre-trained YOLO Model:

- **Strengths:** Fast inference time and high accuracy for common vehicle types.
- **Weaknesses:** Lower precision and recall for vehicle types unique to Indian traffic, leading to reduced overall performance in mixed urban conditions.

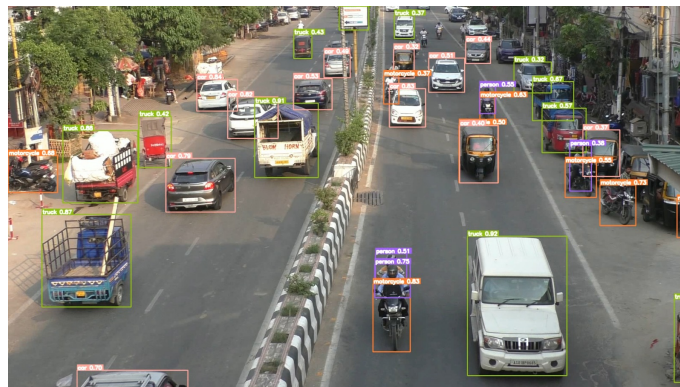


Figure 3.9: Detections and Classifications as per pre-trained model

### 3.1.5 Study Area & Data Collection

The study area selected for this research is a mid-block section of GS Road, specifically near City Center Mall and the Ganeshguri Flyover in Guwahati. This location was chosen due to its high traffic volume and diverse vehicle mix, providing a comprehensive environment to test the YOLO model for traffic analysis.



Figure 3.10: Camara Postion during Data Collection

### 3.1.6 Study Area

#### Location

The study was conducted on a 4-lane divided urban road, GS Road, near City Center Mall and Ganeshguri Flyover, Guwahati.

#### Significance

This area is a major commercial hub with a significant amount of vehicular traffic, making it ideal for studying mixed urban traffic conditions typical of Indian cities.

### 3.1.7 Data Collection

To gather accurate and comprehensive traffic data, a systematic data collection process was implemented using video recordings from strategically positioned cameras.

#### Video Recording Setup

- **Position:** Cameras were installed on a foot overbridge to provide an elevated, unobstructed view of the traffic below.
- **Duration:** Videos were recorded from 8 AM to 5 PM, covering both peak and off-peak traffic hours to ensure a representative sample of traffic conditions throughout the day.
- **Resolution:** The cameras recorded in 1080p resolution at 30 frames per second (fps), providing high-quality footage suitable for detailed analysis.

#### Marking for Real Coordinates

- **Physical Markings:** Cones were used to mark intervals of 10 meters up to 50 meters along the road section. These markings ensured accurate measurement of vehicle speeds and distances.
- **Purpose:** The marked intervals allowed for precise determination of the trap area/length, which is crucial for accurate data collection and subsequent analysis.

## Data Captured

- The video recordings captured various types of vehicles, including cars, buses, motorcycles, non-motorized vehicles (NMCs), pedestrians, trucks, light commercial vehicles (LCVs), and auto-rickshaws.
- The footage was used to extract data on vehicle counts, speeds, and movement patterns within the marked intervals.

### 3.1.8 Conversion in PCU

o Passenger Car Unit (PCU) values are used to standardize the impact of different vehicle types on traffic flow. The number of vehicles was converted into Passenger Car Units (PCU) using PCU factors provided in the Indo-HCM 2017 guidelines.

## 3.2 Deployment of YOLO Model

The trained YOLO model was deployed on recorded videos for vehicle detection, classification, counting, and travel time estimation. This deployment aimed to estimate traffic fundamentals for further analysis and speed compliance evaluation.

However, One notable issue is the misclassification of some vehicle classes, such as Light Commercial Vehicles (LCVs) and buses, particularly when these vehicles are farther from the camera. This problem arises from the smaller dataset available for these classes during training.

### 3.2.1 Enhanced Vehicle Classification

To improve the accuracy of vehicle classification, as this research work is processing recorded video data rather than real-time data, a method was implemented to minimize classification errors. This approach focuses on ensuring that the final classification of each vehicle is as accurate as possible by updating the vehicle class for all previous frames once the vehicle is near the exit of the trap length.

## Technical Implementation

- **Observation:** The model predicts vehicle classes more accurately when vehicles are closer to the camera's viewport.
- **Dictionary Use:** A Python dictionary tracks vehicles, with keys as vehicle tracking IDs and values as lists of frames where each vehicle is within the target region.
- **Class Update:** When a vehicle exits the target region, its class in all previous frames is updated to match the class detected at the exit.
- **Data Storage:** The correct class, along with all its previous frames, is saved in a text file, which is then converted into a CSV file for further analysis.
- **Memory Management:** After recording the data, the corresponding frame list is deleted from the dictionary to ensure efficient memory use and prevent memory overload.

This method ensures that the most accurate class for each vehicle is recorded, significantly improving the reliability of traffic analysis.

### 3.2.2 Setup of Trap Areas

#### Trap Areas

- **Dimensions:** Two trap areas were defined along the width of the road, which is 7.5 meters. The first trap area has a length of 40 meters, and the second has a length of 20 meters. These areas were marked physically during data collection.
- **Cross Section:** A cross section was established near the camera to continuously count vehicles and classify them for traffic flow analysis.

#### Vehicle Tracking

- **Unique ID Assignment:** When vehicles enter the 40-meter trap area, each vehicle is assigned a unique ID, which is used to track the vehicle until it leaves the trap area.

### 3.2.3 Traffic Flow Measurement

#### Definition

- **Traffic flow** is defined as the number of vehicles passing through a given point within a specific period (vehicles per hour, veh/hr).

#### Measurement Process

- Vehicles passing through the cross section near the camera were counted along with their classification over 5-minute intervals.
- The counts were then converted into PCU/5 minutes using the appropriate PCU factors.
- To calculate the traffic flow per hour, the PCU count over 5 minutes was multiplied by 12, yielding PCU/hr.

#### Traffic Density Measurement

1. **Definition:** Traffic density is the number of vehicles occupying a unit length of road at any given instant (vehicles per kilometer, veh/km).

2. **Measurement Process:**

- The number of vehicles within the 40-meter trap length was counted along with their classification in each frame over a 5-minute period.
- The total PCU count for the period was then converted into PCU/km using the formula:

$$\text{Density (PCU/km)} = \left( \frac{\text{PCU count over 5-minute interval} \times 1000}{40 \times 5 \times 60 \times 30} \right)$$

This formula accounts for the length of the trap area, the duration of the interval, and the frame rate of the video.

## Travel Time and Speed

### 1. Travel Time:

- Travel time for each vehicle was recorded from its entry time to its exit time within the 20-meter trap area.
- This data was used to calculate the speed of each vehicle.

2. **Vehicle Speed:** The speed of each vehicle was calculated by dividing the 20-meter distance by the time taken to cross this distance:

$$\text{Speed} = \frac{20 \text{ meters}}{\text{Time taken}}$$

This speed measurement is crucial for further analysis of speed compliance and traffic flow characteristics.

3. **Space Mean Speed:** Space mean speed represents the average speed of vehicles in the traffic stream over a specific roadway segment and time period. It was calculated by taking the average travel time of all vehicles passing through the 20-meter trap length over the 5-minute period:

$$\text{Space Mean Speed} = \frac{20 \text{ meters}}{\text{Average travel time of passing vehicles over 5 min}}$$

## Vehicle Trajectory Analysis

1. **Trajectory Tracking:** Vehicle trajectories were tracked, and their x and y coordinates were recorded along with their unique IDs as they moved through the 40-meter trap length. This data helps in studying movement patterns and behaviors in mixed traffic conditions.

The deployment of the YOLO model facilitated the accurate and efficient collection of essential traffic parameters such as vehicle count, classification, speed, travel time, and density. These metrics are crucial for a detailed understanding and analysis of traffic flow and dynamics in Indian urban mixed traffic conditions, as outlined in the Indo-HCM 2017 guidelines.

# Chapter 4

## ANALYSIS AND OBSERVATIONS

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### 4.1 Analysis and Observations

#### 4.1.1 Traffic Composition

The analysis of traffic composition involves examining the types and proportions of vehicles detected in the traffic stream over a 9-hour period. The following data represents the number and percentage distribution of different vehicle classes observed:

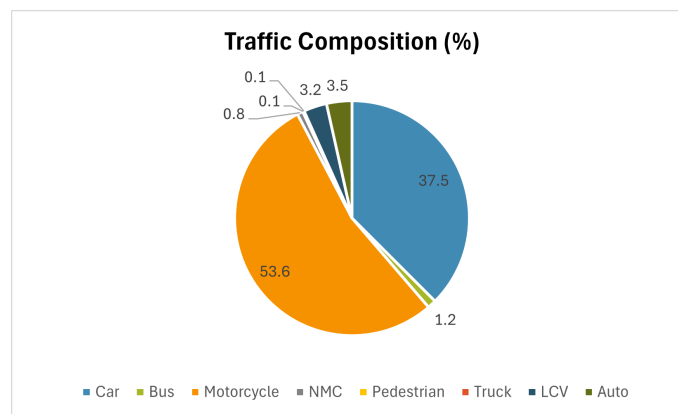


Figure 4.1: Traffic Composition

| Vehicle Type                    | Number | Percentage (%) |
|---------------------------------|--------|----------------|
| Car                             | 12,212 | 37.5           |
| Bus                             | 404    | 1.2            |
| Motorcycle                      | 17,461 | 53.6           |
| Non-Motorized Vehicles (NMV)    | 247    | 0.8            |
| Pedestrian                      | 21     | 0.1            |
| Truck                           | 40     | 0.1            |
| Light Commercial Vehicles (LCV) | 1,059  | 3.3            |
| Auto-rickshaw                   | 1,131  | 3.5            |

Table 4.1: Traffic Composition over a 9-hour Period

The pie chart below visually represents the traffic composition, illustrating the dominance of motorcycles and cars in the traffic stream, followed by smaller proportions of other vehicle types.

## **Analysis and Implications**

The traffic composition reveals several critical insights about urban traffic in developing countries, particularly in Indian cities:

- **Majority of Motorcycles:**

- Motorcycles constitute the majority of the traffic, accounting for 53.6%. This high percentage is typical in developing countries where two-wheelers are a popular mode of transportation due to their affordability and maneuverability.
- The large number of motorcycles significantly influences traffic patterns. Motorcycles often do not adhere strictly to lane discipline, frequently weaving through traffic, exploiting smaller gaps between vehicles, and occasionally overspeeding. This behavior can lead to a more chaotic and less predictable traffic flow, complicating traffic management and safety measures.

- **Significant Presence of Cars:**

- Cars make up 37.5% of the traffic, indicating their substantial presence on the roads. As personal vehicles, they contribute to congestion, especially during peak hours.

- **Notable Percentages of Autos and LCVs:**

- Auto-rickshaws (3.5%) and Light Commercial Vehicles (LCVs) (3.3%) also form a significant portion of the traffic. These vehicles are vital for last-mile connectivity and goods transportation in urban areas.
- The presence of these vehicle types, especially autos, adds to the mixed nature of traffic, as they have different driving characteristics and speeds compared to cars and motorcycles.

- **Lower Percentages of Buses, NMVs, Pedestrians, and Trucks:**

- Buses (1.2%), Non-Motorized Vehicles (NMVs) (0.8%), pedestrians (0.1%), and trucks (0.1%) constitute smaller fractions of the traffic. While their numbers are lower, these vehicle types are crucial for public transportation and logistics.

The traffic composition highlights the unique challenges of managing urban traffic in developing countries like India. The dominance of motorcycles, combined with significant numbers of auto-rickshaws and LCVs, creates a dynamic and complex traffic environment.

#### 4.1.2 Traffic Flow/Volume

Traffic flow, representing the volume of vehicles passing a given point over time, is crucial for understanding road usage dynamics. The following analysis is based on traffic flow data collected over a 9-hour period, with measurements taken at 5-minute intervals and expressed in PCU/hr.

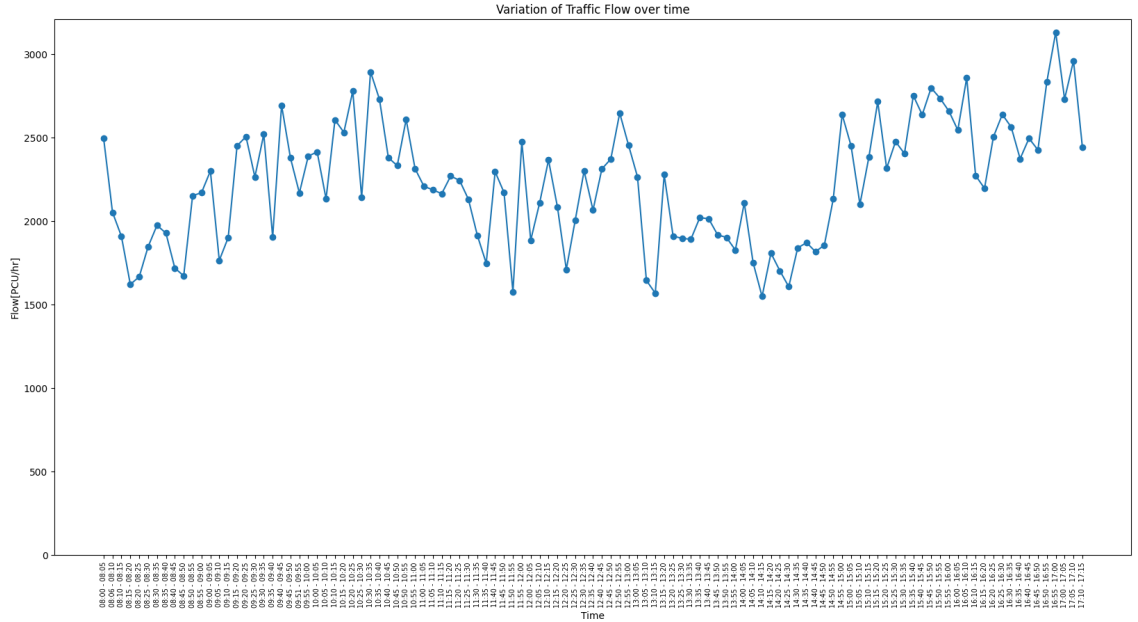


Figure 4.2: Variation of Traffic flow over time

### **Morning Rush Hour (8:00 AM - 10:30 AM)**

- Peak: Noticeable rise in traffic flow, peaking at around 2800 PCU/hr by 10:30 AM.
- Insight: Reflects the morning rush as more vehicles enter the road network.

### **Midday Stability (11:30 AM - 3:00 PM)**

- Stable Flow: Traffic flow stabilizes, fluctuating between 1600 and 2300 PCU/hr.
- Insight: Indicates a balanced traffic volume with fewer surges.

### **Late Afternoon Peak (3:00 PM - 5:00 PM)**

- Significant Rise: Traffic flow increases steadily, reaching up to 3400 PCU/hr towards 5:00 PM.
- Insight: Indicates the evening rush hour as people start heading home from work.

Traffic flow increases significantly towards late afternoon, peaking just before 5:00 PM, indicating the evening rush hour.

### 4.1.3 Traffic Density

Assessment of traffic density in various road segments and during different times of the day is crucial for understanding road usage patterns. The graph below presents the variation of traffic density over time.

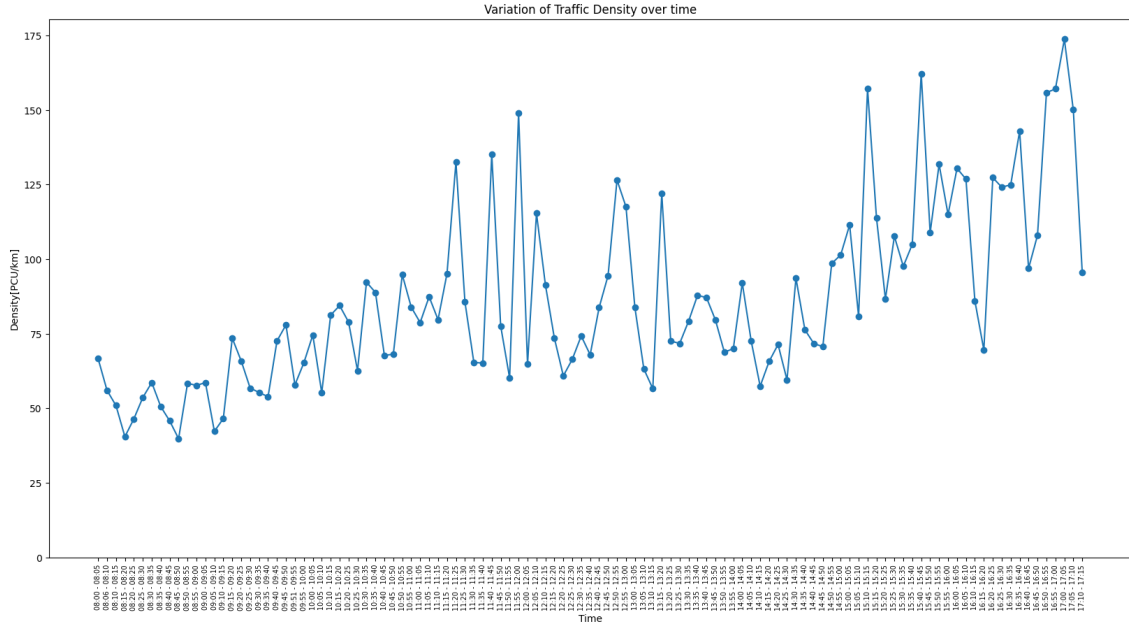


Figure 4.3: Variation of Traffic density over time

Noticeable rise in traffic density, peaking at around 100 PCU/km by 10:30 AM. Traffic density stabilizes, fluctuating between 70 and 120 PCU/km. Traffic density increases steadily, reaching up to 160 PCU/km towards 5:00 PM.

These significant trends highlight the morning rush hour peak, midday stability, late afternoon to evening peak, and the impact of sudden spikes due to illegal parking or stopping, which are crucial for effective traffic management and planning.

### 4.1.4 Space Mean Speed

Calculation of the average speed of vehicles in the traffic stream provides crucial insights into traffic flow dynamics. The graph below presents the variation of space mean speed over time.

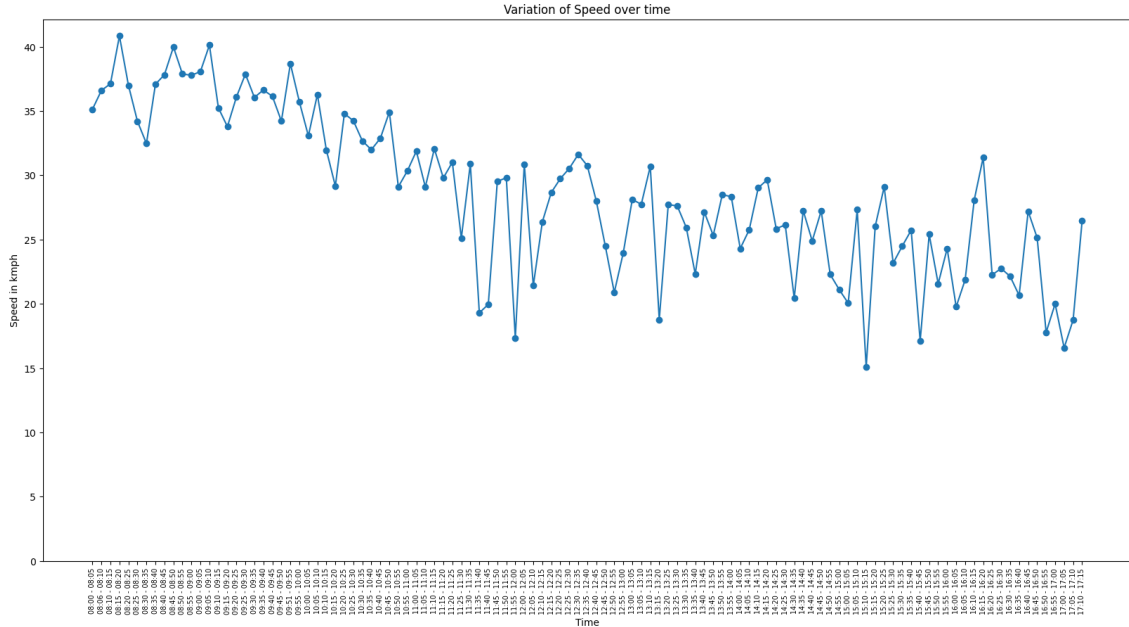


Figure 4.4: Variation of Speed over time

- Noticeable decline in speed, reducing from 35-40 km/h in the morning to around 30 km/h by 10:30 AM.
- Speed stabilizes, fluctuating between 25 and 35 km/h during midday.
- Speeds show significant variability in the afternoon, ranging from 20 to 30 km/h.
- Speed decreases further, averaging around 20-25 km/h towards 5:00 PM.

These significant trends highlight the morning speed decline, midday stability, afternoon variability, and late afternoon decrease, which are crucial for traffic analysis.

### 4.1.5 Greenshields Model Fitting

The Greenshields model was applied to understand the relationship between traffic density, speed, and flow. This model is a fundamental traffic flow theory that relates these parameters through linear relationships.

#### Fitting Process

- **Model Assumptions:**

- Linear Relationship: Greenshields model assumes a linear relationship between speed ( $V$ ) and density ( $K$ ).
- Equations:
  - \* Speed-Density Relationship:  $V = V_f \left(1 - \frac{K}{K_j}\right)$
  - \* Flow-Density Relationship:  $Q = K \cdot V$
  - \* Combining the above, Flow-Speed Relationship:  $Q = K_j \cdot V_f \cdot \left(\frac{V}{V_f}\right) \cdot \left(1 - \frac{V}{V_f}\right)$

- **Model Fitting:**

- Jam Density ( $K_j$ ): The density at which speed drops to zero.
- Free-Flow Speed ( $V_f$ ): The speed at which density is zero.
- Optimal Density ( $K_{opt}$ ): The density at which the maximum flow occurs.
- Optimal Speed ( $V_{opt}$ ): The speed at which the maximum flow occurs.
- Optimal Flow ( $Q_{opt}$ ): The maximum flow rate.

- **Estimation of Parameters:**

- Jam Density ( $K_j$ ): Derived from the linear speed-density relationship where speed equals zero.
- Free-Flow Speed ( $V_f$ ): Estimated as the maximum observed speed in free-flow conditions.

- Optimal Density ( $K_{opt}$ ) and Optimal Speed ( $V_{opt}$ ): Calculated from the model, where optimal density is half of jam density ( $K_{opt} = \frac{K_j}{2}$ ) and optimal speed is half of free-flow speed ( $V_{opt} = \frac{V_f}{2}$ ).
- Optimal Flow ( $Q_{opt}$ ): Determined using the relationship  $Q_{opt} = K_{opt} \cdot V_{opt}$ .

## Results and Graphs

### • Speed vs. Density (V-K)

- **Observation:** Speed decreases linearly with increasing density, from a free-flow speed of 43.1 km/h to zero at the jam density (254.6 PCU/km).

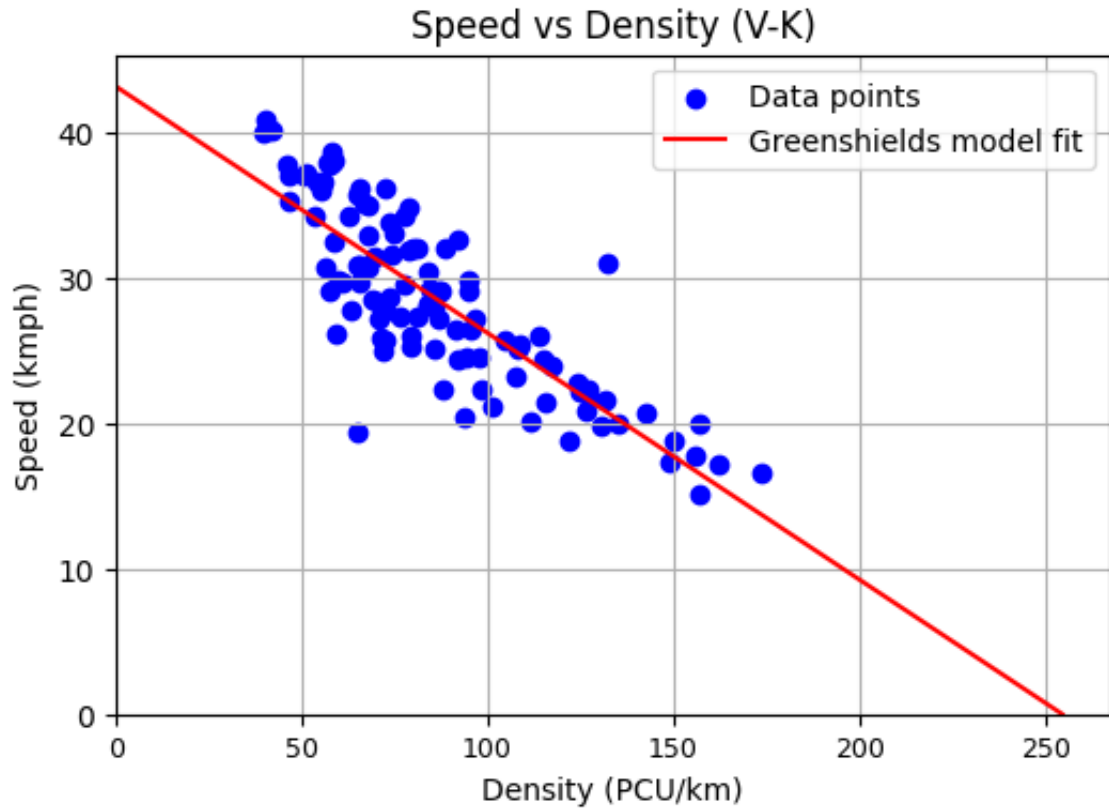


Figure 4.5: Speed vs. Density (V-K)

- **Flow vs. Density (Q-K)**

- **Observation:** Flow increases with density up to the optimal density (127.3 PCU/km), then decreases, peaking at 2745.1 PCU/hr.

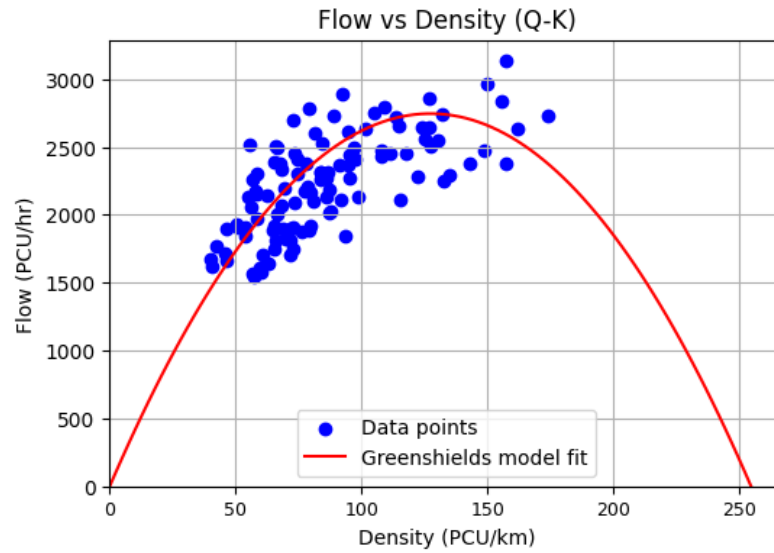


Figure 4.6: Speed vs. Density (V-K)

- **Speed vs. Flow (V-Q)**

- **Observation:** Speed decreases as flow increases up to the optimal flow, beyond which both speed and flow decrease.

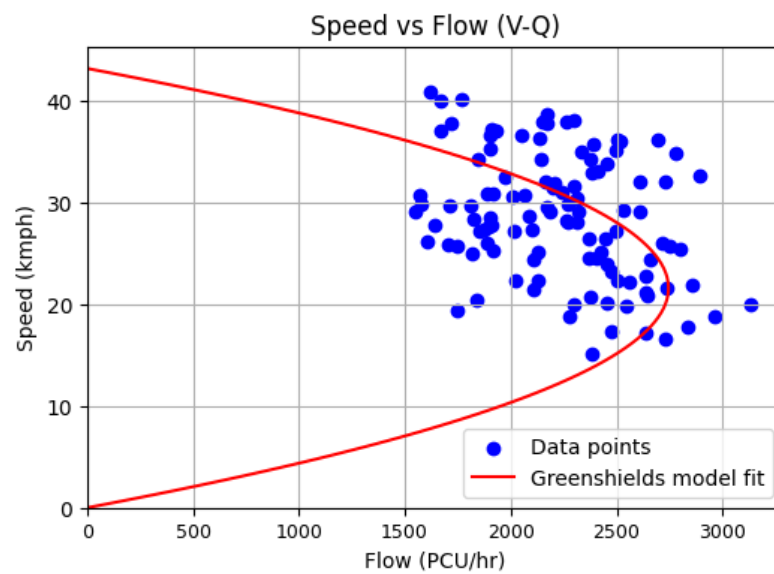


Figure 4.7: Speed vs. Flow (V-Q)

- Jam Density ( $K_j$ ): 254.59688537657192 PCU/km
- Optimal Density ( $K_{opt}$ ): 127.29844268828596 PCU/km
- Optimal Speed ( $V_{opt}$ ): 21.564222768574098 kmph
- Free-flow Speed ( $V_f$ ): 43.128445537148195 kmph
- Optimal Flow ( $Q_{opt}$ ): 2745.091976222761 PCU/hr

#### 4.1.6 Level of Service

Level of Service (LOS) is a qualitative measure used to describe the operational conditions of a roadway segment and the perception of these conditions by motorists and passengers. It reflects factors such as speed, travel time, traffic interruptions, and comfort. LOS is categorized into six levels, from A to F, with A representing the best operating conditions and F the worst (Indo-HCM 2017).

| LOS   | Volume/Capacity Ratio | Percentage of Free Flow Speed |
|-------|-----------------------|-------------------------------|
| LOS A | $\leq 0.15$           | $\geq 84$                     |
| LOS B | 0.15 – 0.45           | 83 – 76                       |
| LOS C | 0.46 – 0.75           | 75 – 59                       |
| LOS D | 0.76 – 0.85           | 58 – 41                       |
| LOS E | 0.86 – 1.00           | 40 – 22                       |
| LOS F | $> 1.00$              | $< 22$                        |

Table 4.2: LOS of Multilane Divided Urban Roads based on Stream Speed, V/C Ratio and FFS

Evaluation of the Level of Service (LOS) for different road segments is critical for assessing traffic performance and planning improvements. LOS is determined using two methods: the flow/capacity (V/C) ratio and the percentage of free flow speed (FFS).

## Flow/Capacity (V/C) Ratio

Higher V/C ratios indicate higher levels of congestion.

- **Flow vs. Density (Q-K):** The graph shows a parabolic relationship, with flow increasing to an optimal point before decreasing as density increases. Using the V/C ratio in the observed data, LOS is found as C, D, E, and F.

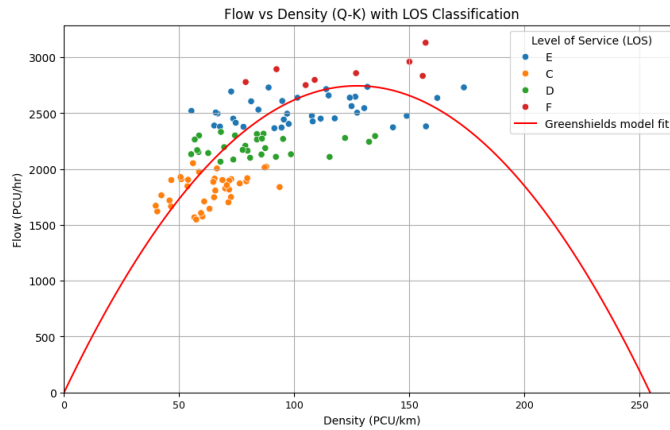


Figure 4.8: Flow vs. Density (Q-K) with LOS Classification

- **Speed vs. Flow (V-Q):** Speed decreases as flow increases, following the Greenshields model. Higher flow rates with lower speeds result in lower LOS grades.

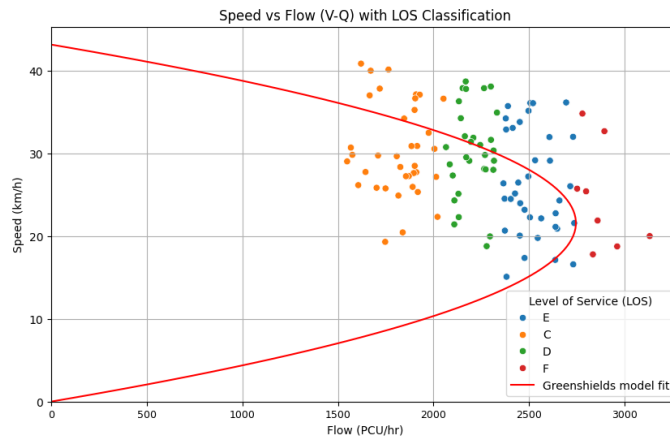


Figure 4.9: Speed vs. Flow (V-Q) with LOS Classification

The potential undervaluation of capacity in theoretical models suggests that practical road capacity may exceed the values indicated by Indo-HCM 2017, reinforcing the need for empirical validation.

### 4.1.7 Speed Compliance

The analysis of speed compliance involves comparing the detected vehicle speeds with the posted speed limits on the road. The attached graphs illustrate the number of vehicles from different classes that did not adhere to the speed limits.

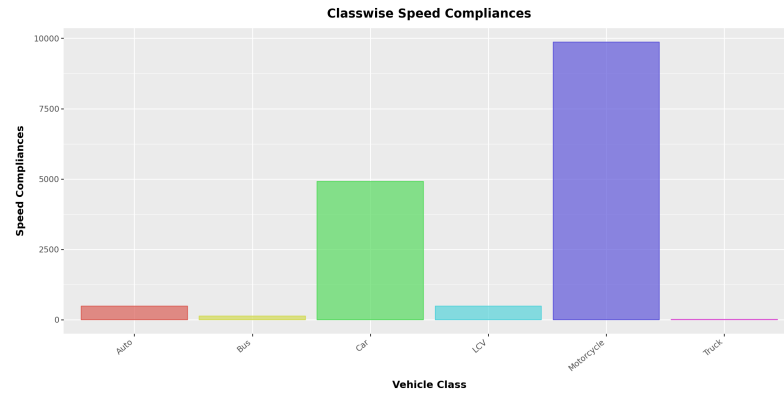


Figure 4.10: Class Wise Speed Compliances

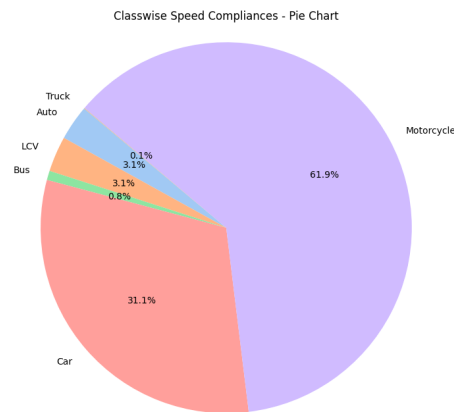


Figure 4.11: Class Wise Speed Compliances

- Motorcycles have the highest rate of speed non-compliance, reflecting a common trend in developing countries where motorcycles often do not adhere to lane discipline and tend to over-speed.
- A significant proportion of cars also exceed the speed limit, which might be due to the relatively higher speed capabilities of these vehicles.
- Other vehicle classes such as LCVs, buses, autos, and trucks show much lower rates of non-compliance, possibly due to their lower speed capabilities and stricter regulations.

## Hourly Speed Non-Compliance

- The highest number of speed violations occur between 10:00 AM and 12:00 PM, with peaks at 2594 and 2649 non-compliant instances respectively.
- There is a noticeable decline in speed violations after 12:00 PM, continuing to decrease steadily throughout the afternoon.

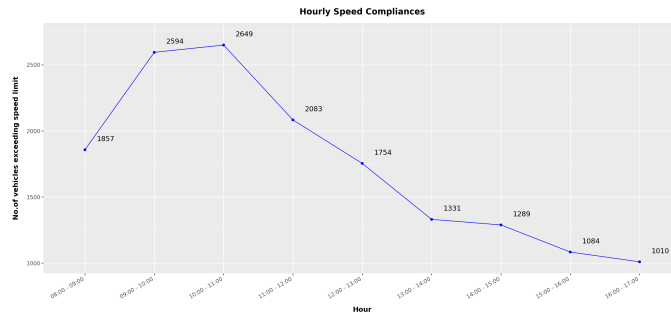


Figure 4.12: Class Wise Speed Compliances

## Class wise Speed Non-Compliance

- Motorcycles consistently show the highest level of non-compliance across all hours, followed by cars, with other vehicle classes exhibiting significantly lower numbers.

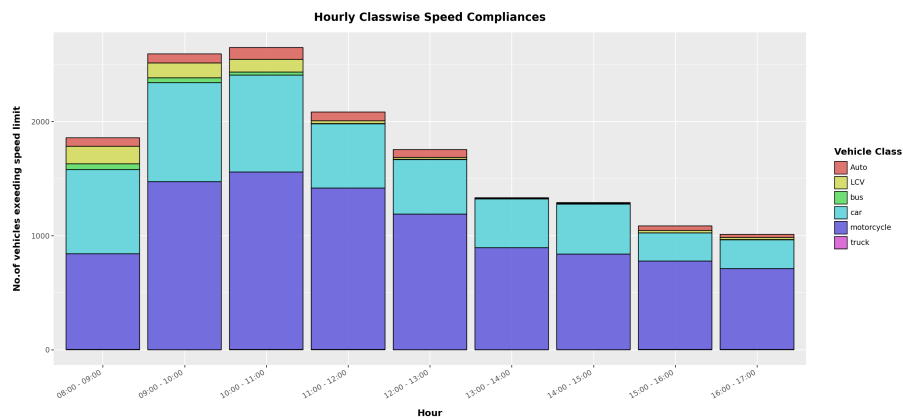


Figure 4.13: hourly classwise speed compliances

# Chapter 5

## CONCLUSION AND FUTURE SCOPE

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### 5.1 Conclusion

The research presented in this thesis has successfully developed and implemented a deep learning-enabled vehicle detection and traffic analysis system tailored for the complex conditions of Indian urban mixed traffic. By leveraging the YOLOv8 model, the study effectively addressed the challenges posed by heterogeneous traffic, lack of lane discipline, and varying vehicle types and sizes. The model performs well within a 50-60 meter range, but some misclassifications and occasional undetection of vehicles still occur. To address these limitations and enhance the model's accuracy and reliability in traffic analysis and speed compliance evaluations, it is essential to augment the training dataset with more diverse and comprehensive images. This includes increasing the number of samples for underrepresented classes and incorporating more side-view images. By doing so, the model's performance will improve, particularly in detecting vehicles from side views, which is currently hindered by the limited side-view data in the training set. Key findings from the study include:

- **Enhanced Vehicle Detection:** The YOLOv8 model demonstrated a high accuracy rate in detecting and classifying different types of vehicles under diverse traffic conditions. This capability is crucial for real-time traffic monitoring and management.
- **Traffic Flow and Density Analysis:** The system provided detailed insights into traffic flow and density, enabling a better understanding of traffic patterns and congestion levels. This data is vital for urban planners and traffic management authorities to devise strategies to mitigate traffic congestion.
- **Speed Compliance Monitoring:** The analysis of speed compliance highlighted significant variations in adherence to speed limits across different vehicle types and times of the day. Motorcycles showed the highest level of non-compliance, pointing to the need for targeted interventions.

## 5.2 Future Scope

This study has significantly contributed to traffic analysis in mixed urban environments, but several areas offer potential for further research and enhancement:

- **Integration with Traffic Systems:** Integrate the developed system with existing traffic management infrastructure, including traffic signal control and incident management systems, for a comprehensive traffic management solution.
- **Scalability and Generalization:** Expand the study to cover diverse geographical areas with different traffic conditions to validate the robustness and scalability of the models, ensuring reliability in varied environments.
- **Vehicle Trajectory Analysis:** Improve trajectory accuracy and smoothness by addressing bounding box fluctuations and using advanced tracking algorithms. Conduct further studies on trajectory prediction to understand vehicle movement patterns in mixed traffic.
- **Real-time Deployment:** Develop and test the model for real-time traffic monitoring and analysis, providing immediate benefits for traffic management and enforcement. Integrate with existing systems for automated monitoring and control.
- **Comprehensive Traffic Management:** Combine vehicle detection with other traffic parameters, such as pedestrian movement and traffic signal timings, for a holistic traffic management approach. Explore applications for accident detection and emergency response.

By addressing these areas, future research can enhance the capabilities and applications of deep learning in traffic analysis and management.

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